

**[William Exclusive] Foundational Arithmetic Recovery & Powers
Mastery (Set 1)**

Name:

William

Target:

 Focus: Decimal Arithmetic / Index Form Expansion /
3-Step Pipeline

Time: 35 Total: 100 Score:

Mins

Pts



Teacher's Key Reminders (Read aloud before starting):

- ① **Powers are NOT Multiplications:** When you see an exponent a^b , you MUST write it out as repeated multiplication! $2^3 = 2 \times 2 \times 2 = 8$ (Writing 6 is strictly prohibited!).
- ② **Decimal Addition Alignment:** $0.9 + 0.1 = 1.0$. Ten tenths make one whole number in the ones place!
- ③ **Decimal Division:** In $5.4 \div 0.6$, multiply both numbers by 10 to get $54 \div 6 = 9$.
- ④ **Three-Step Template:** Never stop after finding Y . You must finish Step 2 to solve for X before submitting!

Section 1: Daily 5-Minute Decimal Arithmetic Sprint (2.5 pts each, 30 pts total) Score:

Observe the operators and decimal points carefully. Write your answers directly (Time limit: 6 mins):

(1) $0.9 + 0.1 =$

(2) $3.8 + 0.2 =$

(3) $1 - 0.25 =$

(4) $5 - 0.8 =$

(5) $5.4 \div 0.6 =$

(6) $30 \div 0.5 =$

(7) $0.36 \div 0.09 =$

(8) $2.5 \times 0.4 =$

(9) $1.2 \times 0.5 =$

(10) $0.7 \times 0.8 =$

(11) $4.2 \div 0.7 =$

(12) $100 \times 0.035 =$

Section 2: Powers as Repeated Multiplication (25 pts total) Score:

1. [Mandatory Expanded Form Table] (1 pt per blank, 15 pts total)

⚠ You MUST write the expanded repeated multiplication in the middle column!

Index Form	Expanded Form (Repeated Multiplication)	Final Evaluated Value
2^1	2	2
2^2	2×2	4
2^3	$2 \times 2 \times 2$	8 (⚠ NOT 6!)
2^4		
2^5		
3^2		
3^3		
3^4		
4^3		
5^3		

2. [Powers vs Multiplication Comparison] Fill in each blank with ">", "<", or "=": (2 pts each, 6 pts total)

$$2^3 \text{ ______ } 2 \times 3$$

$$3^4 \text{ ______ } 3 \times 4$$

$$5^2 \text{ ______ } 5 \times 2$$

3. [Product of Powers Rule] Simplify $10^4 \times 10^5$. (4 pts)

Hint: 10^4 has four 10s multiplied; 10^5 has five 10s multiplied. How many 10s are multiplied in total? Add or multiply the exponents?

$$10^4 \times 10^5 = 10^{\text{---} + \text{---}} = 10^{\text{---}}.$$

(⚠ Warning: Exponents are added! Never multiply them to get $10^{4 \times 5} = 10^{20}$!)

Section 3: Three-Step Structured Problem Solving (12.5 pts each, 25 pts total)

Score: _____

 The Three-Step Pipeline Standard:

Step 1 Circle given values and write the governing rule/formula.

Step 2 Solve systematically step-by-step; check off each variable as found.

Step 3 State the complete conclusion clearly. Finish to the end!

4. [Six-Digit Divisibility Problem · Three-Step Full Solution]

The six-digit number $\overline{91X93Y}$ is divisible by both 8 and 9. Write the complete solution:

Step 1 Use the last 3 digits to find Y:

Because the number is divisible by 8, its last 3 digits $\overline{93Y}$ must be divisible by 8.
 $930 \div 8 = 116$ remainder 2.

So $20 + Y$ must be a multiple of 8. For a single digit Y , $20 + Y = 24$,

Therefore: $Y =$ _____ (✓ Y is solved!)

Step 2 Use the sum of digits to find X (William's Focus: DO NOT STOP HERE!):

Now the six-digit number is $\overline{91X93Y}$.

Because the number is divisible by 9, the sum of all digits must be a multiple of 9.

Sum of digits $= 9 + 1 + X + 9 + 3 + () =$ _____

Since $X \in \{0, 1, \dots, 9\}$, for the sum to be divisible by 9,

Therefore: $X =$ _____ (✓ X is solved!)

Step 3 State the final answer:

Answer: $X =$ _____, $Y =$ _____, the six-digit number is _____.

5. [Bilingual Word Problem · Speed and Distance]

Leo cycles at an average speed of 15 km/h. He rides for 40 minutes. How far does he ride?

Step 1 Identify units and convert:

Speed is in km/h, but time is given as 40 minutes. You MUST convert minutes to hours!

Time $t = 40$ minutes $= \frac{40}{60}$ hours $= \frac{=}{=}$ hours.

Step 2 Apply the distance formula:

Formula: Distance $=$ Speed $\times$ Time

$S = 15 \times \frac{=}{=}$ = _____ km.

Step 3 Conclude with units:

Answer: Leo rides _____ km.

Section 4: Smart Fraction Comparison (5 pts each, 20 pts total)

Score: _____

6. Fill in each blank with ">" or "<", and write a one-sentence reason:

(a) $\frac{7}{12}$ _____ $\frac{7}{15}$

Reason (Same numerators):

(b) $\frac{5}{9}$ _____ $\frac{7}{9}$

Reason (Same denominators):

(c) $\frac{13}{15}$ _____ $\frac{17}{19}$

Reason (Distance to 1 benchmark):

(d) $\frac{5}{8}$ _____ $\frac{7}{12}$

Reason (Common denominator 24):



[William Set 1] Answer Key & Scoring Guide (For Review)

Section 1: Daily Decimal Sprint (30 pts, 2.5 pts each)

- (1) 1.0 (or 1) (2) 4.0 (or 4) (3) 0.75 (4) 4.2 (5) 9 (⚠ NOT 0.9!) (6) 60
 (7) 4 (8) 1.0 (or 1) (9) 0.6 (10) 0.56 (11) 6 (12) 3.5

Section 2: Powers as Repeated Multiplication (25 pts)

1. (14 pts)

- $2^4 = 2 \times 2 \times 2 \times 2 = 16$

- $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$

- $3^2 = 3 \times 3 = 9$

- $3^3 = 3 \times 3 \times 3 = 27$

- $3^4 = 3 \times 3 \times 3 \times 3 = 81$ (⚠ NOT 18!)

- $4^3 = 4 \times 4 \times 4 = 64$

- $5^3 = 5 \times 5 \times 5 = 125$

2. (6 pts) $2^3 > 2 \times 3$ ($8 > 6$); $3^4 > 3 \times 4$ ($81 > 12$); $5^2 > 5 \times 2$ ($25 > 10$).

3. (5 pts) $10^4 \times 10^5 = 10^{4+5} = 10^9$.

Section 3: Three-Step Structured Solving (25 pts)

4. (12.5 pts)

Step 1: $Y = 6$ (4 pts);

Step 2: Sum of digits $= 9 + 1 + X + 9 + 3 + 6 = 28 + X$. For $28 + X$ to be divisible by 9, $X = 8$ (5.5 pts);

Step 3: Answer: $X = 8$, $Y = 6$, number is 918936 (3 pts).

5. (12.5 pts)

Step 1: Time $t = \frac{40}{60} = \frac{2}{3}$ hours (4 pts);

Step 2: $S = 15 \times \frac{2}{3} = 10$ km (5.5 pts);

Step 3: Answer: 10 km (3 pts).

Section 4: Smart Fraction Comparison (20 pts, 5 pts each)

(a) $>$. Reason: With identical numerators, smaller denominators represent larger parts.

(b) $<$. Reason: With identical denominators, greater numerators indicate greater values.

(c) $<$. Reason: $1 - \frac{13}{15} = \frac{2}{15}$, $1 - \frac{17}{19} = \frac{2}{19}$. Since $\frac{2}{15} > \frac{2}{19}$, the distance to 1 is larger, so the value is smaller.

(d) $>$. Reason: With common denominator 24, $\frac{5}{8} = \frac{15}{24}$ and $\frac{7}{12} = \frac{14}{24}$. Since $15 > 14$, $\frac{5}{8} > \frac{7}{12}$.